Cocompletion under sifted colimits need not be finitely accessible

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Slides (PDF)

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1 Background

2 Counterexamples

Finitely presentable objects

$\mathscr{C}$: a category with filtered colimits.

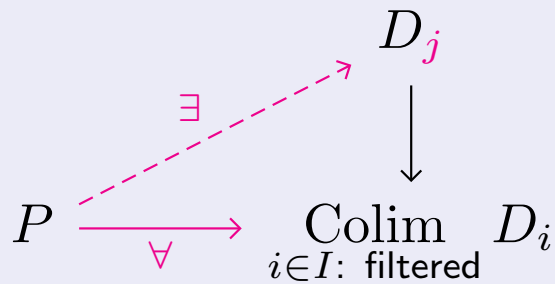
Definition

$P \in \mathscr{C}$ is finitely presentable (f.p.) $\overset{\text{def}}{\Leftrightarrow}$ $\mathscr{C}(P, -)$ preserves filtered colimits.

$$\operatorname*{Colim}_i \mathscr{C}(P, D_i) \cong \mathscr{C}(P, \operatorname*{Colim}_i D_i)$$

Remark

P: finitely presentable $\Longrightarrow$



for some j

Example

$P \in \mathbf{Set}, \mathbf{Grp}, \mathbf{Ring}$, is f.p.

$\Longleftrightarrow$ P is generated by finite elements modulo finite (finitary) equations.

Finitely accessible categories

Definition

$\mathscr{C}$ is finitely accessible $\overset{\text{def}}{\Leftrightarrow}$

1. $\mathscr{C}$ has filtered colimits;
2. $\exists\ \mathscr{P} \subseteq \mathscr{C}$: small full sub. s.t.
   1. $\mathscr{P} \subseteq \{\text{finitely presentables}\}$;
   2. Every object in $\mathscr{C}$ can be written as a filtered colimit of objects from $\mathscr{P}$.

Example

- $\mathbf{Set}$, $\mathbf{Grp}$, $\mathbf{Ring}$: the categories of sets, groups, rings
- $\mathbf{Pos}$: — of posets
- $\mathbf{Cat}$: — of small categories
- $\mathbf{Graph}$: — of graphs ⚠ in graph theorists' terminology
- *Explore more on CatDat*

Non-example

- $\mathbf{Top}$: — of topological spaces
- *Explore more on CatDat*

Finitely accessible categories with limits

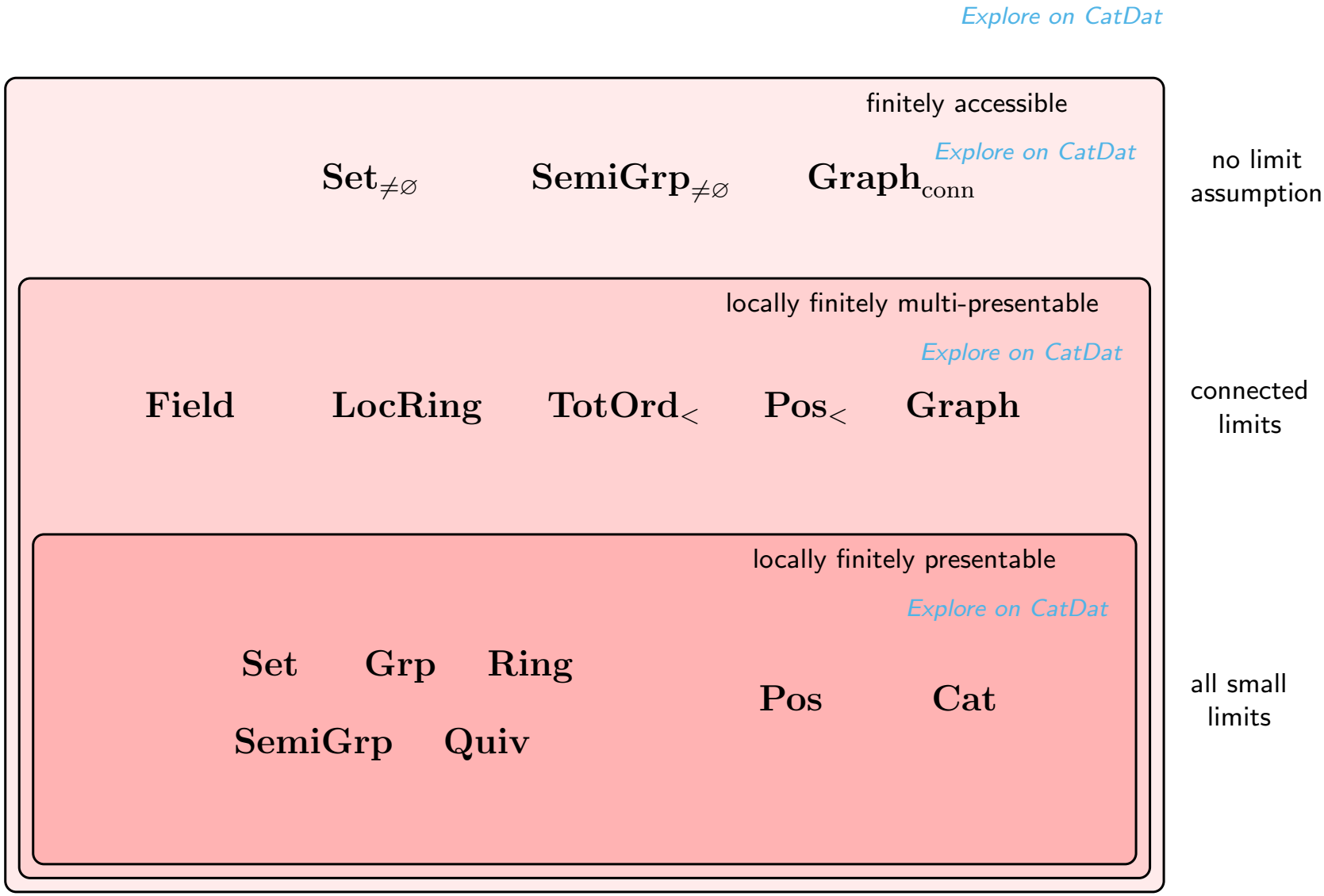
Definition [Die80a; AR94]

$$\begin{aligned} \text{locally finitely multi-presentable} &:= \text{finitely accessible} + \text{connected limits} \\ &\iff \text{finitely accessible} + \text{multi-colimits} \\ \text{locally finitely presentable} &:= \text{finitely accessible} + \text{limits} \\ &\iff \text{finitely accessible} + \text{colimits} \end{aligned}$$

Fact [AR94, 4.32, 1.52]

1. $\mathscr{C}$: l.f.m-p. $\iff$ $\mathscr{C} \simeq \mathbf{Mod}\,\mathbb{S}$ ($\exists\mathbb{S}$: finite-limit coproduct sketch)
   cf. essentially multi-algebraic theory [AR96], disjunctive theory [Joh79].
2. $\mathscr{C}$: l.f.p. $\iff$ $\mathscr{C} \simeq \mathbf{Mod}\,\mathbb{S}$ ($\exists\mathbb{S}$: finite-limit sketch)
   cf. essentially algebraic theory [Fre72], cartesian theory [Joh02], partial Horn theory [PV07].

Classification



Filtered and sifted colimits

$\mathscr{I}$: a small category.

Definition

$$\mathscr{I}\text{: filtered} \overset{\text{def}}{\Leftrightarrow} \mathscr{I}\text{-colimits commute with finite limits in } \mathbf{Set}.$$

$$\mathscr{I}\text{: sifted} \overset{\text{def}}{\Leftrightarrow} \mathscr{I}\text{-colimits commute with finite products in } \mathbf{Set}.$$

Remark

$$\text{filtered} \subsetneq \text{sifted}$$

Example

The walking reflexive pair is sifted, but not filtered:

$$X \underset{d_1}{\overset{d_0}{\underset{\longleftarrow s \longrightarrow}{\longrightarrow}}} Y \qquad d_0 s = d_1 s = \mathrm{id}_Y.$$

Colimits of this shape are called *reflexive coequalizers*.

Replacing "filtered" with "sifted" I

~~Recall~~ Definition [AR01]

$P \in \mathscr{C}$ is ~~finitely presentable (f.p.)~~ (strongly finitely presentable (s.f.p.)) $\overset{\text{def}}{\Leftrightarrow}$ $\mathscr{C}(P, -)$ preserves ~~filtered~~ (sifted) colimits.

~~Recall~~ Definition [AR01]

$\mathscr{C}$ is ~~finitely accessible~~ (a generalized variety) $\overset{\text{def}}{\Leftrightarrow}$

1. $\mathscr{C}$ has ~~filtered~~ (sifted) colimits;
2. $\exists\ \mathscr{P} \subseteq \mathscr{C}$: small full sub. s.t.
   1. $\mathscr{P} \subseteq \{$ ~~finitely presentables~~ (strongly finitely presentables) $\}$;
   2. Every object in $\mathscr{C}$ can be written as a ~~filtered~~ (sifted) colimit of objects from $\mathscr{P}$.

Replacing "filtered" with "sifted" II

~~Recall~~ Definition [Die80b; AR01]

multi-algebraic — generalized variety

~~locally finitely multi-presentable~~ $:=$ ~~finitely accessible~~ + connected limits

$\iff$ ~~finitely accessible~~ + multi-colimits

finitary algebraic — generalized variety

~~locally finitely presentable~~ $:=$ ~~finitely accessible~~ + limits

$\iff$ ~~finitely accessible~~ + colimits

~~Recall~~ Fact [AR01, 4.4][ARV11, 6.9]

1. $\mathscr{C}$: ~~l.f.m-p.~~ (m-alg.) $\iff$ $\mathscr{C} \simeq \mathbf{Mod}\,\mathbb{S}$ ($\exists\mathbb{S}$: ~~finite-limit~~ coproduct sketch) (finite-product)

   cf. multi-algebraic theory [Die80b].

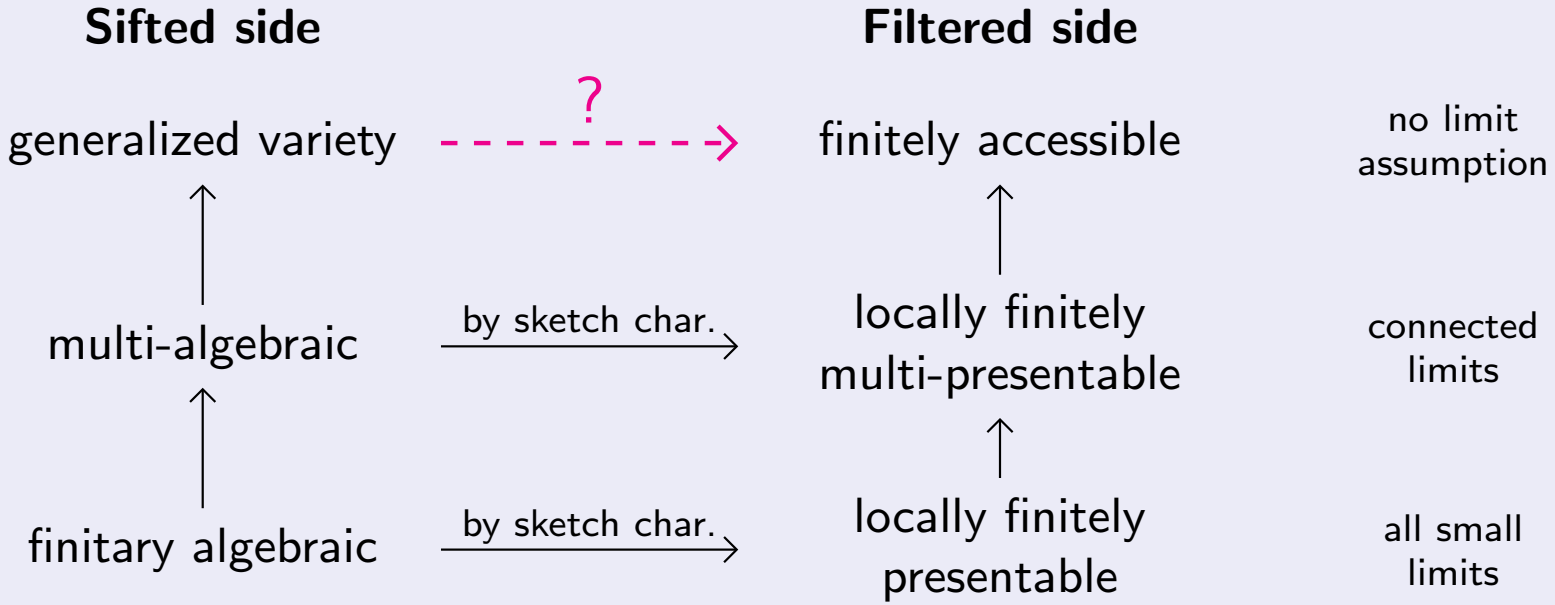
2. $\mathscr{C}$: ~~l.f.p.~~ (f.alg.) $\iff$ $\mathscr{C} \simeq \mathbf{Mod}\,\mathbb{S}$ ($\exists\mathbb{S}$: ~~finite-limit~~ sketch) (finite-product)

   cf. equational theory, Lawvere theory.

Sifted side $\longrightarrow$ Filtered side

Fact

Each arrow exhibits the implication:

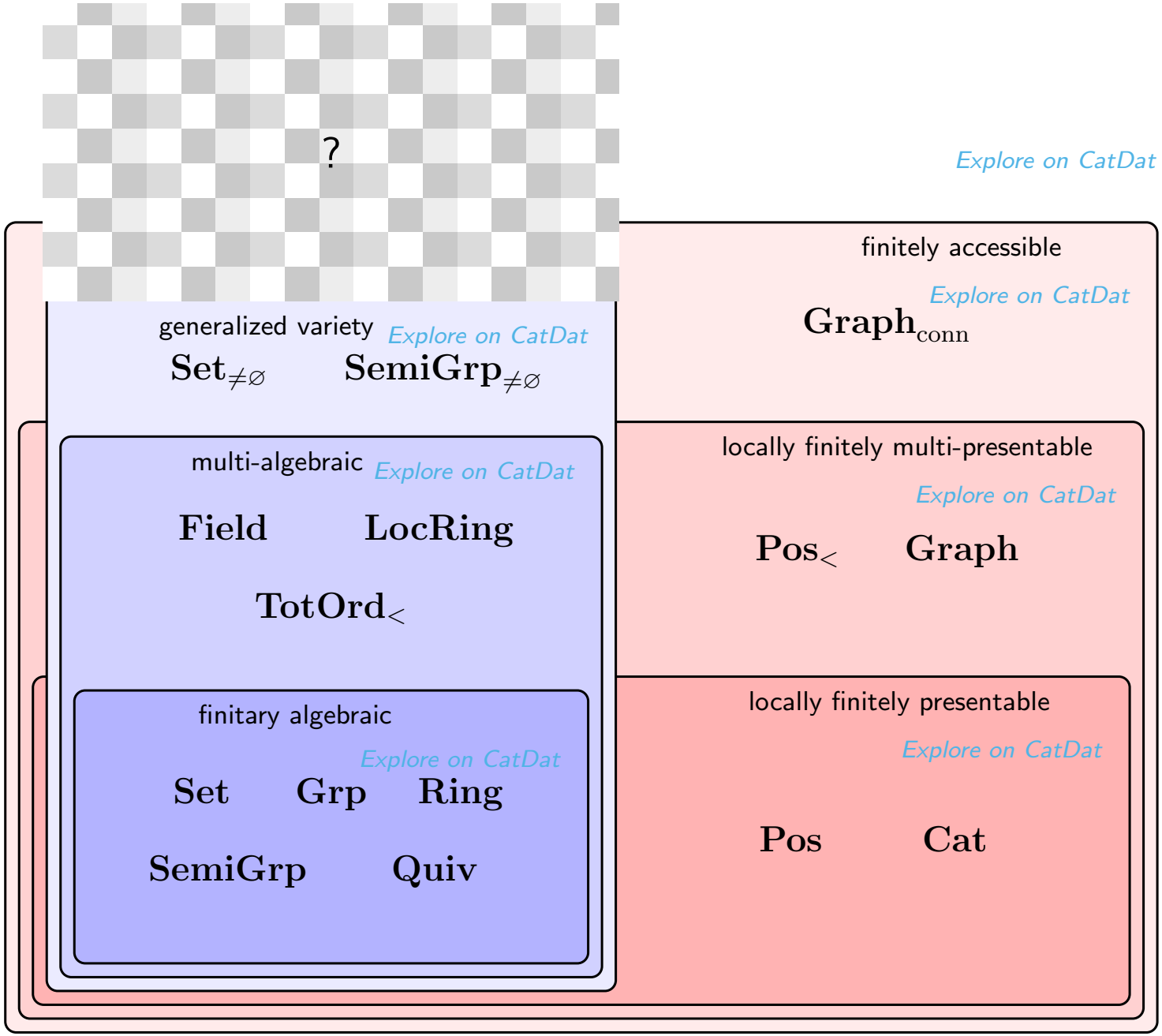


Question [AR01, 4.9(1)]

$$\text{generalized variety} \overset{?}{\Longrightarrow} \text{finitely accessible}$$

Remark [AR01, 4.8(2)]; [Ten25, 4.11(2)]

"generalized variety $\Longrightarrow$ $\aleph_1$-accessible" was already known.


?
Explore on CatDat
finitely accessible
Explore on CatDat
$\mathbf{Graph}_{\mathrm{conn}}$
no limit assumption
generalized variety
Explore on CatDat
$\mathbf{Set}_{\neq\emptyset}$
$\mathbf{SemiGrp}_{\neq\emptyset}$
multi-algebraic
Explore on CatDat
$\mathbf{Field}$
$\mathbf{LocRing}$
$\mathbf{TotOrd}_{<}$
locally finitely multi-presentable
Explore on CatDat
$\mathbf{Pos}_{<}$
$\mathbf{Graph}$
connected limits
finitary algebraic
Explore on CatDat
$\mathbf{Set}$
$\mathbf{Grp}$
$\mathbf{Ring}$
$\mathbf{SemiGrp}$
$\mathbf{Quiv}$
locally finitely presentable
Explore on CatDat
$\mathbf{Pos}$
$\mathbf{Cat}$
all small limits

The cocompletion formulation

$\mathscr{A}$: small. $\quad \mathbf{Psh}(\mathscr{A}) := \mathbf{Set}^{\mathscr{A}^{\mathrm{op}}}$.

Notation

$$\begin{aligned} \mathbf{Ind}(\mathscr{A}) &:= \{P \in \mathbf{Psh}(\mathscr{A}) \mid \textstyle\int P \text{ is filtered}\} \\ &= \text{the filtered-colimit cocompletion of } \mathscr{A}. \\ \mathbf{Sind}(\mathscr{A}) &:= \{P \in \mathbf{Psh}(\mathscr{A}) \mid \textstyle\int P \text{ is sifted}\} \\ &= \text{the sifted-colimit cocompletion of } \mathscr{A}. \end{aligned}$$

The Yoneda embedding is decomposed as follows:

$$\mathscr{A} \xhookrightarrow{\text{よ}} \mathbf{Ind}(\mathscr{A}) \hookrightarrow \mathbf{Sind}(\mathscr{A}) \hookrightarrow \mathbf{Psh}(\mathscr{A})$$

Fact

$$\begin{aligned} \mathscr{C}\text{: finitely accessible} &\iff \mathscr{C} \simeq \mathbf{Ind}(\mathscr{A}) \text{ for some small } \mathscr{A}. \\ \mathscr{C}\text{: generalized variety} &\iff \mathscr{C} \simeq \mathbf{Sind}(\mathscr{A}) \text{ for some small } \mathscr{A}. \end{aligned}$$

Question [AR01, 4.9(1)] (rephrased)

Is $\mathbf{Sind}(\mathscr{A})$ finitely accessible for any small $\mathscr{A}$?

Positive cases

$\mathscr{A}$: small.

Positive case 1

$\mathscr{A}$: essentially finite $\Longrightarrow$ $\mathbf{Sind}(\mathscr{A})$: essentially finite $\wedge$ finitely accessible.

Positive case 2 (the *Ind–Rec decomposition*)

- $\mathscr{A}$ has finite coproducts [AR01, 2.3(2)][ARV11, 7.4] or
- $\mathscr{A}$ has pullbacks [Che21, Thm. 5.1]

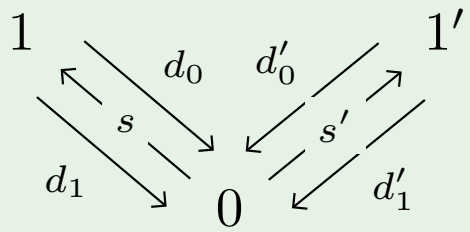
$\Longrightarrow$ $\mathbf{Sind}(\mathscr{A}) \simeq \mathbf{Ind}\big(\mathbf{Rec}(\mathscr{A})\big)$ $\Longrightarrow$ $\mathbf{Sind}(\mathscr{A})$: finitely accessible.

$\mathbf{Rec}(\mathscr{A})$: the reflexive-coequalizer cocompletion.

Failure of the Ind–Rec decomposition

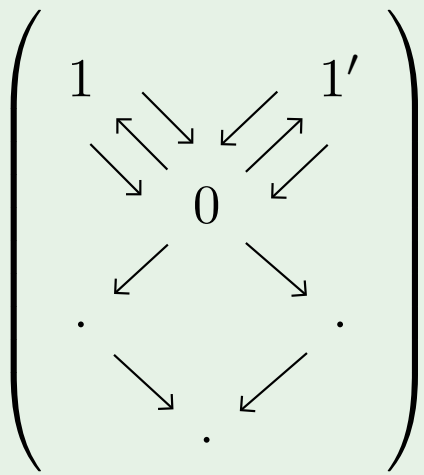
Example

$\mathscr{D}$: the walking twin reflexive pairs.

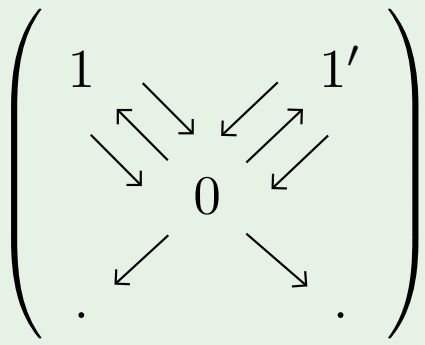


$$d_i s = d_i' s' = \mathsf{id}_0.$$

$$\mathbf{Sind}(\mathscr{D}) \simeq$$



$$\mathbf{Ind}(\mathbf{Rec}(\mathscr{D})) \simeq \mathbf{Rec}(\mathscr{D}) \simeq$$



$\rightsquigarrow\!\!\!\rightarrow$ $\mathbf{Sind}(\mathscr{D}) \not\simeq \mathbf{Ind}(\mathbf{Rec}(\mathscr{D}))$, but $\mathbf{Sind}(\mathscr{D})$ is finitely accessible ($\because \mathscr{D}$: finite)

1 Background

2 Counterexamples

The negative answer

Theorem [Kaw26]

$\exists \mathscr{A}$: small s.t. $\mathbf{Sind}(\mathscr{A})$ is not finitely accessible.

Corollary

generalized variety $\not\Longrightarrow$ finitely accessible.

The category $\mathscr{A} = \mathscr{A}(\mathscr{S})$

$\mathscr{S}$: small, sifted, not filtered (e.g. the walking reflexive pair).

Construction (the category $\mathscr{A} = \mathscr{A}(\mathscr{S})$)

Object $\mathrm{Obj}\mathscr{A} := \mathrm{Obj}\mathscr{S} \times \underset{\text{ranks}}{\mathbb{N}}$;

Non-identity morphism

$$(X, \underset{\text{rank}}{m}) \xrightarrow{(s, \underset{\text{label}}{F})} (Y, \underset{\text{rank}}{n}), \text{ where } \begin{array}{c} m < n, \quad X \xrightarrow{s} Y \quad \text{in } \mathscr{S}, \\ \varnothing \neq F \in \mathcal{P}_{\mathrm{fin}}(\mathbb{N}) \text{ s.t. } n \leq \max F; \end{array}$$

Identity morphism $\mathsf{id}_{(X,m)} := (\mathsf{id}_X, \varnothing)$;

Composition $(s_2, F_2) \circ (s_1, F_1) := (s_2 \circ s_1, F_2 \cup F_1)$.

Remark

In $\mathscr{A}$, any non-identity morphism strictly increases the *rank* of objects.
(i.e., $\mathscr{A}$ is a *direct category*).

The presheaves P_U

$U \subseteq \mathbb{N}$: infinite.

Construction

The wide subcategory $\mathscr{A}_U \subseteq \mathscr{A}$: same objects; arrows (s, F) with $F \subseteq U$.

$$\text{The presheaf } P_U := \mathrm{Colim}\big(\mathscr{A}_U \overset{\text{incl.}}{\to} \mathscr{A} \overset{\text{よ}}{\hookrightarrow} \mathbf{Psh}(\mathscr{A})\big).$$

Explicitly, $P_U(A) = \{A \xrightarrow{f} \cdot\}/\!\sim_U$, $\quad f \sim_U k \circ f'$ for k in $\mathscr{A}_U$.

Proposition

1. $\mathscr{A}_U$ is sifted. $\quad \therefore P_U \in \mathbf{Sind}(\mathscr{A})$.
2. $\int P_U$ is not filtered. $\quad \therefore P_U \notin \mathbf{Ind}(\mathscr{A})$.
3. $P_{\mathbb{N}} \cong 1$, the terminal presheaf.
4. $V = \bigcup_m U_m$, $U_m \subseteq U_{m+1} \implies P_V \cong \mathrm{Colim}_m P_{U_m}$ in $\mathbf{Psh}(\mathscr{A})$.

Proof sketch.

1. Use the siftedness of $\mathscr{S}$.
2. An "incoequalizable" pair in $\mathscr{S}$ lifts to one in $\int P_U$.
3. Because $\mathscr{A}_{\mathbb{N}} = \mathscr{A}$.
4. Since the label F is finite for any mor. $f = (s, F)$ in $\mathscr{A}$, $F \subseteq V \iff \exists m.\ F \subseteq U_m$. □

The label detectors Q_U

$U \subseteq \mathbb{N}$: infinite.

Construction

Define $Q_U \in \mathbf{Psh}(\mathscr{A})$ as follows:

$$Q_U(A) \coloneqq \mathcal{P}_{\text{fin}}(\mathbb{N} \setminus U), \quad Q_U(f) \coloneqq (F \setminus U) \cup -, \quad f = (s, F).$$

Lemma

There is a natural transformation $P_U \longrightarrow Q_U$.

Proof. Postcomposition by a morphism in $\mathscr{A}_U$ leaves the label outside U unchanged. Hence the maps

$$P_U(A) \longrightarrow Q_U(A), \qquad [f]_U \mapsto F \setminus U \quad \text{for } f = (s, F)$$

are well-defined. Naturality follows since composition takes unions of labels. □

The orthogonality lemma

The following is a combinatorial core:

Lemma

$U, V \subseteq \mathbb{N}$: infinite, $U \cap V$: finite, $P \in \mathbf{Sind}(\mathscr{A})$,
$P_U \leftarrow P \rightarrow P_V \quad \Longrightarrow \quad P$: representable.

Proof. We have $Q_U \overset{\alpha}{\leftarrow} P \overset{\beta}{\rightarrow} Q_V$. Fix $(A, x) \in \int P$. For $(A, x) \xrightarrow{f=(s,F)} (B, y)$,

$$\begin{array}{ccc} P(A) & \xleftarrow{P(f)} & P(B) \\ & x \longleftarrow\!\shortmid\; y & \\ {\scriptstyle \alpha_A}\downarrow & \downarrow \quad \downarrow & \downarrow{\scriptstyle \alpha_B} \\ \alpha_A(x) & = \quad (F \setminus U) \cup \alpha_B(y) \longleftarrow\!\shortmid\; & \alpha_B(y) \\ Q_U(A) & \xleftarrow[Q_U(f)]{} & Q_U(B) \end{array}$$

$\therefore\ F \setminus U \subseteq \alpha_A(x)$,
Similarly, $F \setminus V \subseteq \beta_A(x)$.

$$H := (U \cap V) \cup \alpha_A(x) \cup \beta_A(x) \quad \Longrightarrow \quad F \subseteq H.$$

H: finite and independent of $f \rightsquigarrow$ the rank of B is bounded by H.

$\int P$: sifted $\rightsquigarrow$ $\int P$ has a terminal object $\rightsquigarrow$ P: representable. $\square$

Two orthogonal chains and the conclusion

$E \coloneqq \{\text{even numbers}\}$, $O \coloneqq \{\text{odd numbers}\}$.

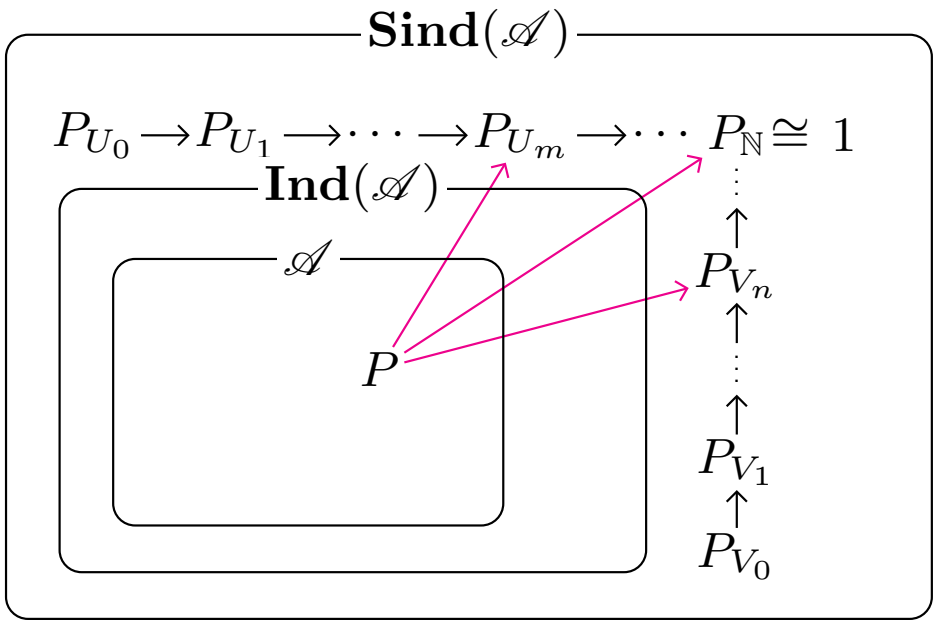
$$U_m \coloneqq E \cup \{2k+1 \mid 0 \leq k < m\}, \qquad V_n \coloneqq \{2k \mid 0 \leq k < n\} \cup O.$$

$\bigcup_m U_m = \bigcup_n V_n = \mathbb{N}$.
$\rightsquigarrow \operatorname{Colim}_m P_{U_m} \cong \operatorname{Colim}_n P_{V_n} \cong P_{\mathbb{N}}\ (\cong 1)$.

$P \in \mathbf{Sind}(\mathscr{A})$: f.p.
$\rightsquigarrow P_{U_m} \leftarrow P \rightarrow P_{V_n}\ (\exists m, n)$.

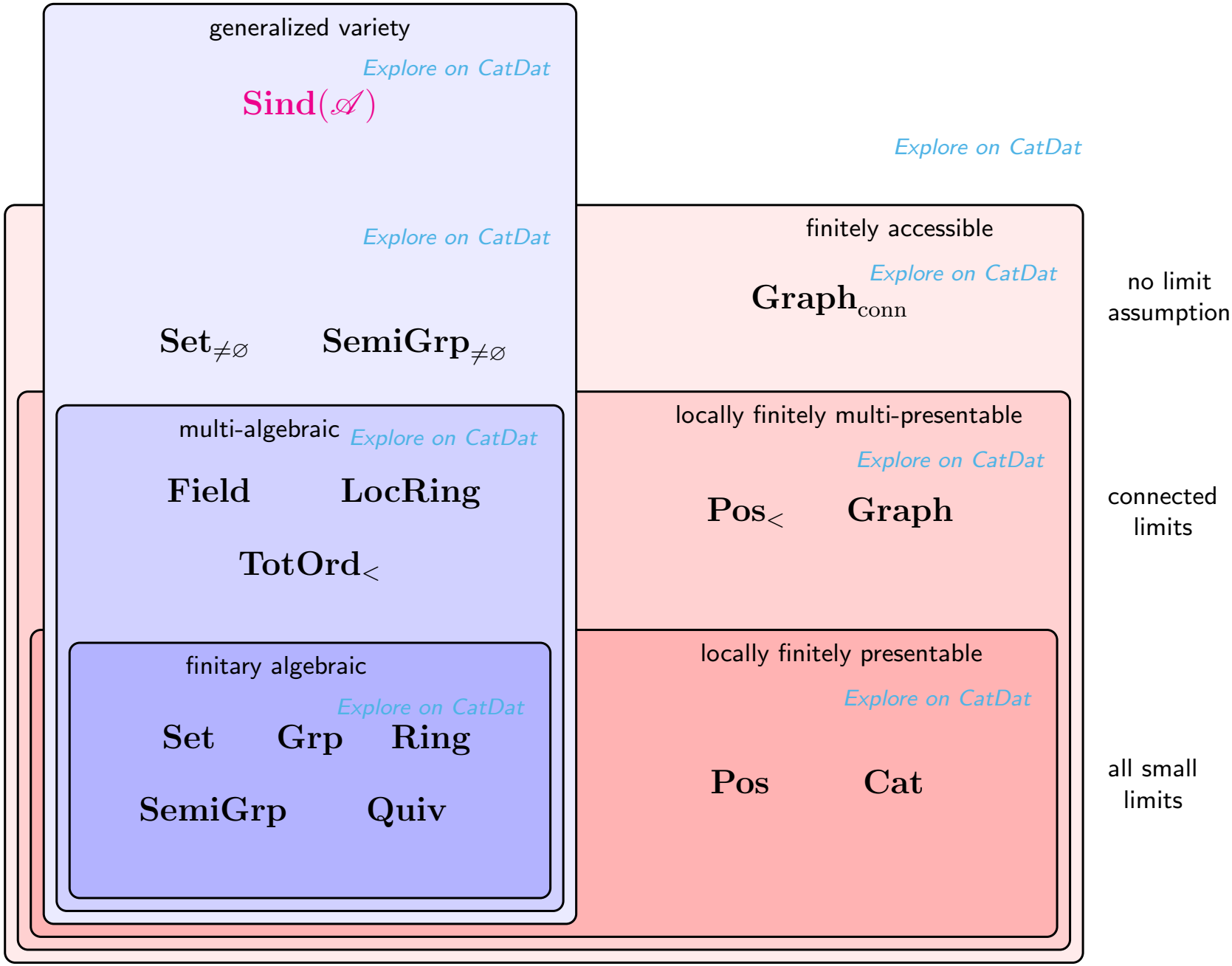
$U_m \cap V_n$: finite
$\rightsquigarrow P$: representable.

But, $\mathbf{Ind}(\mathscr{A})$: filtered-colimit closed



Conclusion

$\mathbf{Sind}(\mathscr{A})$ is not finitely accessible.


generalized variety
Explore on CatDat
$\mathbf{Sind}(\mathscr{A})$
Explore on CatDat
Explore on CatDat
finitely accessible
Explore on CatDat
$\mathbf{Graph}_{\mathrm{conn}}$
$\mathbf{Set}_{\neq\varnothing}$
$\mathbf{SemiGrp}_{\neq\varnothing}$
multi-algebraic
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$\mathbf{Cat}$
$\mathbf{SemiGrp}$
$\mathbf{Quiv}$
no limit assumption
connected limits
all small limits

Thank you!

Preprint

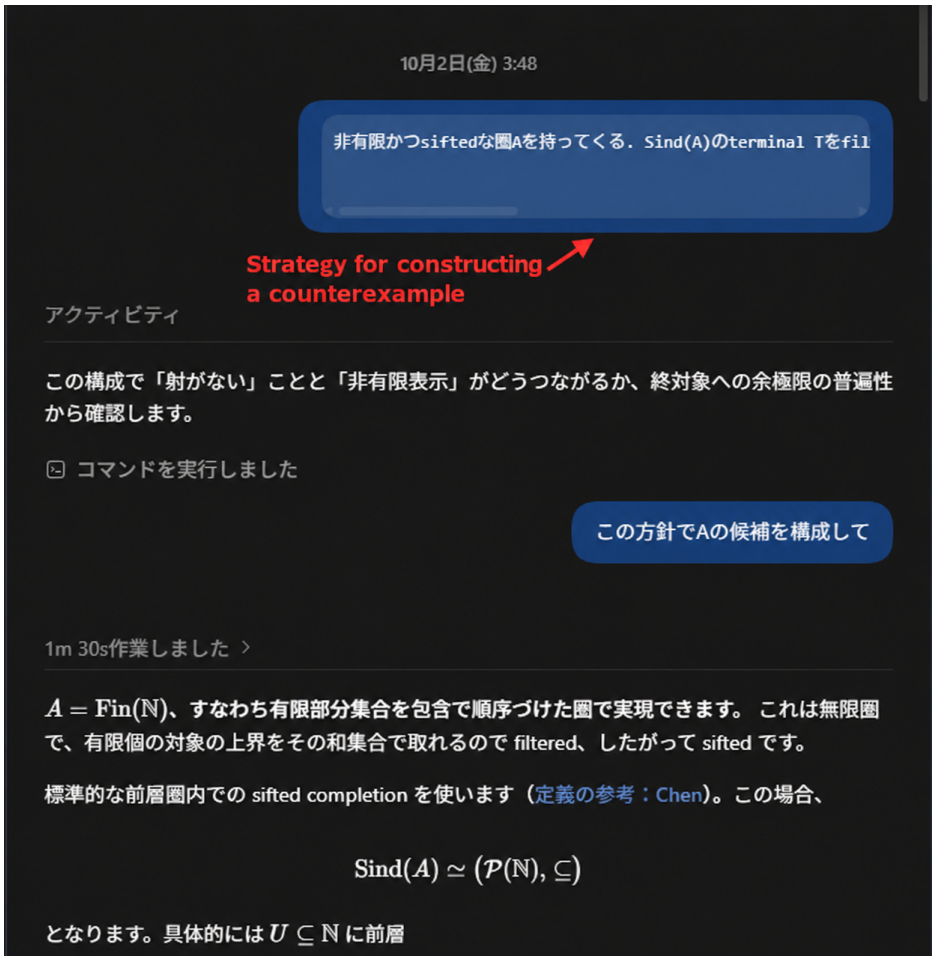
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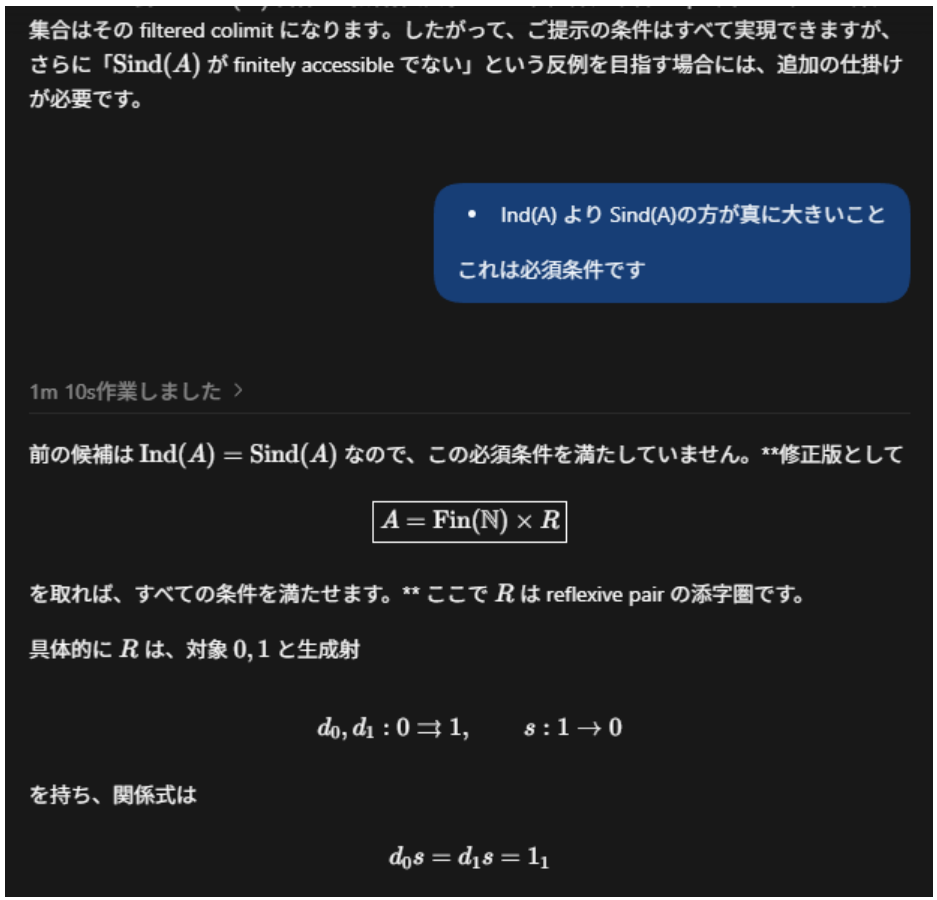
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Slides (PDF)

AI disclosure: 1/2





AI disclosure: 2/2

したがって X_n が fp なら

$$\{*\} = \mathrm{Hom}(X_n, T) \cong \operatorname*{colim}_m \mathrm{Hom}(X_n, Y_m) = \varnothing$$

という矛盾が生じます。Y_m も対称的に非 fp です。

Tをω-chain colimitで各別の方法が簡単にみつかってしまうので、これでは反例にならない

5m 24s作業しました ›

その通りです。前の A は、有限アクセシビリティの反例候補として撤回します。

実際、有限な $F \subseteq \mathbb{N}$ に対して

$$Q_F \cong \mathrm{coeq}\left(y(F,0) \underset{y(1_F,d_1)}{\overset{y(1_F,d_0)}{\rightrightarrows}} y(F,1) \right)$$

は表現可能前層の有限余極限なので fp です。したがって

$$T = \operatorname*{colim}_{n<\omega} Q_{\{0,\dots,n\}}$$

という fp 対象による表示が残ります。

引き続き A を探索して

36m 45s作業しました ›

新しい候補が得られました。今回は、$\mathrm{Sind}(A)$ の fp 対象がすべて表現可能になる、という証明案まで付きます。 独立検証前の構成ですが、前の例にあった別の fp 近似を排除する議論です。

鍵は、射に付けた有限集合のラベルを、合成するときにも残すことです。

R を walking reflexive pair

$$d_0, d_1 : 0 \rightrightarrows 1, \qquad s : 1 \to 0, \qquad d_0 s = d_1 s = 1_1$$

とします。R は sifted ですが、d_0, d_1 を等化する後続射はありません。

正整数の集合を $C = \{1, 2, \dots\}$ とし、A を次で定義します：

$$\mathrm{Ob}(A) = \{(r, n) : r \in \{0,1\},\ n \in \mathbb{N}_0\}.$$

$n < m$ のとき

$$A((r,n),(t,m)) = \{(f,U) : f \in R(r,t),\ \varnothing \neq U \subseteq C \text{ finite},\ m \leq \max U\}.$$

同じ段数では恒等射だけを許し、段数を下げる射はありません。合成は

$$(g,V) \circ (f,U) = (gf, U \cup V).$$

Categories in the classification diagrams

Symbol	Objects	Morphisms
$\mathbf{Set}$	sets	maps
$\mathbf{Set}_{\neq\varnothing}$	non-empty sets	〃
$\mathbf{Grp}$	groups	homomorphisms
$\mathbf{Ring}$	unital rings (not necessarily commutative)	〃
$\mathbf{Field}$	(commutative) fields	〃
$\mathbf{SemiGrp}$	semigroups	〃
$\mathbf{SemiGrp}_{\neq\varnothing}$	non-empty semigroups	〃
$\mathbf{LocRing}$	commutative local rings	local homomorphisms
$\mathbf{Pos}$	posets	monotone maps: $x \leq y \implies f(x) \leq f(y)$
$\mathbf{Pos}_{<}$	posets	strictly monotone maps: $x < y \implies f(x) < f(y)$
$\mathbf{TotOrd}_{<}$	totally ordered sets	strictly monotone maps
$\mathbf{Graph}$	simple undirected graphs without loops	adjacency-preserving maps on vertices
$\mathbf{Graph}_{\mathrm{conn}}$	connected ones in $\mathbf{Graph}$	adjacency-preserving maps on vertices
$\mathbf{Quiv}$	quivers (directed graph)	natural transformations
$\mathbf{Cat}$	small categories	functors

References

[AR01] J. Adámek and J. Rosický. "On sifted colimits and generalized varieties". In: *Theory Appl. Categ.* 8 (2001), pp. 33–53.

[ARV11] J. Adámek, J. Rosický, and E. M. Vitale. *Algebraic Theories*. Cambridge Univ. Press, 2011.

[AR94] J. Adámek and J. Rosický. *Locally Presentable and Accessible Categories*. Cambridge Univ. Press, 1994.

[AR96] J. Adámek and J. Rosický. "An algebraic description of locally multipresentable categories". In: *Theory Appl. Categ.* 2 (1996), pp. 40–54.

[CatDat] *CatDat*. https://catdat.app.

[Che21] R. Chen. "On sifted colimits in the presence of pullbacks". In: *Theory Appl. Categ.* 37 (2021), pp. 1176–1193.

[Die80a] Y. Diers. "Catégories localement multiprésentables". In: *Arch. Math. (Basel)* 34.4 (1980), pp. 344–356.

[Die80b] Y. Diers. "Catégories multialgébriques". In: *Arch. Math. (Basel)* 34.3 (1980), pp. 193–209.

[Fre72] P. Freyd. "Aspect of topoi". In: *Bull. Austral. Math. Soc.* 7 (1972), pp. 1–76.

[Joh79] P. T. Johnstone. "A syntactic approach to Diers' localizable categories". In: *Applications of Sheaves*. Vol. 753. Lecture Notes in Math. Springer, 1979, pp. 466–478.

[Joh02] P. T. Johnstone. *Sketches of an Elephant: A Topos Theory Compendium. Vol. 2*. Oxford Univ. Press, 2002.

[Kaw26] Y. Kawase. *Cocompletion under sifted colimits need not be finitely accessible*. 2026. arXiv: 2610.08548 [math.CT].

[PV07] E. Palmgren and S. J. Vickers. "Partial Horn logic and cartesian categories". In: *Ann. Pure Appl. Logic* 145.3 (2007), pp. 314–353.

[Ten25] G. Tendas. "More on soundness in the enriched context". In: *J. Pure Appl. Algebra* 229.11, 108110 (2025).